The documentation and process conversion measures necessary to comply with this revision shall be completed by 20 January 1994

INCH POUND

MIL-S-19500/539A
20 October 1993
SUPERSEDING
MIL-S-19500/539(USAF)
30 April 1980

MILITARY SPECIFICATION

SEMICONDUCTOR DEVICE, DARLINGTON TRANSISTOR, NPN, SILICON, POWER
TYPES 2N6300, 2N6301 JANTX AND JANTXV

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for NPN, Darlington, silicon, power transistors. Two levels of product assurance are provided for each device type as specified in MIL-S-19500. For JAN quality assurance level (see 6.3).

1.2 Physical dimensions. See figure 1, similar to TO-3.

1.3 Maximum ratings.

	P_T 1/		V_{CBO}	V_{CEO}	V_{EBO}	I_C	I_B	T_{OP} and T_{STG}
	T_C = 0°C	T_C = 100°C						
	W	W	V dc	V dc	V dc	A dc	mA dc	°C
2N6300	75	32	60	60	5	8	120	-55 TO +200
2N6301	75	32	80	80	5	8	120	-55 TO +200

1/ Derate linearly at 0.428 W/°C above T_C > 0°C.

1.4 Primary electrical characteristics.

	h_{FE2} 1/ V_{CE} = 3 V dc I_C = 4 A dc	h_{FE3} 1/ V_{CE} = 3 V dc I_C = 8 A dc	$\|h_{fe}\|$ V_{CE} = 3 V dc I_C = 3 A dc f = 1 MHz	C_{obo} 100 kHz ≤ f ≤ 1 MHz V_{CB} = 10 V dc I_E = 0	Pulse response	
					t_{on}	t_{off}
				pF	µs	µs
Min	750	100	25			
Max	18000		350	200	2.0	8.0

See footnote at end of table.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: NASA/Parts Project Office (NPPO) NASA Goddard Space Flight Center, Code 310.A, Greenbelt, MD 20771 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A

FSC 5961

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1.4 Primary electrical characteristics. Continued.

	$V_{BE(sat)}$ I_C = 8 A dc I_B = 80 mA dc 1/	$V_{CE(sat)}$ I_C = 8 A dc I_B = 80 mA dc 1/	h_{fe} V_{CE} = 3 V dc I_C = 3 A dc f = 1 kHz	$R_{\theta JC}$
	V dc	V dc		°C/W
Min			300	
Max	4.0	3.0		2.33

1/ Pulsed see 4.5.1

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

MILITARY

MIL-S-19500 - Semiconductor Devices, General Specification for.

STANDARD

MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Associated detail specification. The individual item requirements shall be in accordance with MIL-S-19500, and as specified herein.

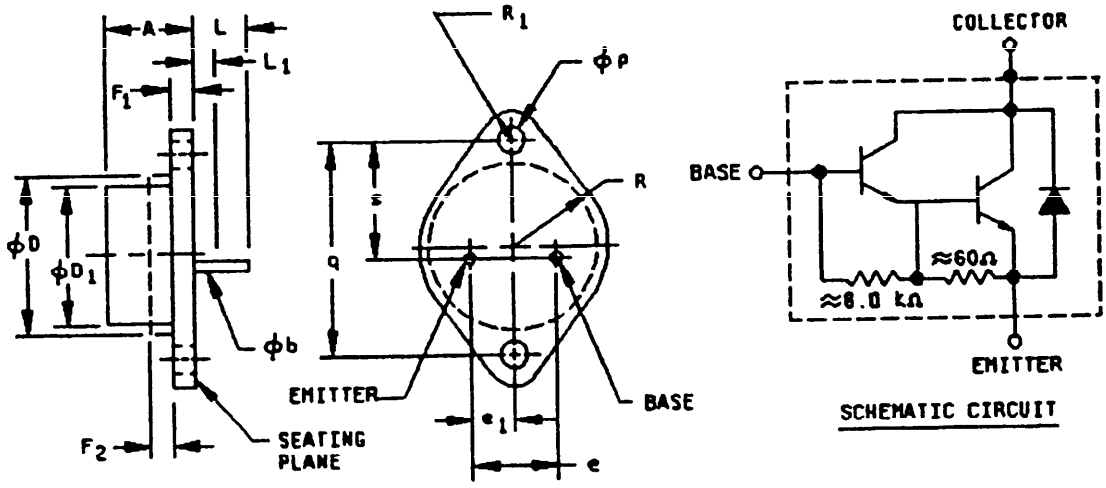
3.2 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-S-19500.

3.3 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-S-19500, and on figure 1 herein.

3.3.1 Lead finish. Lead finish shall be solderable in accordance with MIL-S-19500, MIL-STD 750, and as specified herein. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

3.4 Marking. Marking shall be in accordance with MIL-S-19500.

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Ltr	Dimensions				Notes	Ltr	Dimensions				Notes
	Inches		Millimeters				Inches		Millimeters		
	Min	Max	Min	Max			Min	Max	Min	Max	
A	.250	.340	6.35	8.64		L	.360	.500	9.14	12.70	5, 9
ϕb	.028	.034	0.71	.86	4, 5, 9	L_1		.050		1.27	5, 9
ϕD		.620		15.75	3	ϕp	.142	.152	3.61	3.86	7
ϕD_1	.470	.500	11.94	12.70	3	q	.958	.962	24.33	24.43	
e	.190	.210	4.83	5.33	4	R		.350		8.89	
e_1	.093	.107	2.36	2.72	3	R_1	.115	.145	2.92	3.68	6
F_1	.050	.075	1.27	1.91	3	s	.570	.590	14.48	14.99	4
F_2		.050		1.27	3						

FIGURE 1. Physical dimensions.

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NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Body contour is optional within zone defined by ϕD, ϕD_1.
4. Measured at points .050 (1.27 mm) - .055 (1.4 mm) below the seating plane. When gauge is not used, measurement will be made at the seating plane.
5. Both terminals.
6. At both ends.
7. Two holes.
8. The collector shall be electrically connected to the case.
9. ϕb applies between L_1 and L. Diameter is uncontrolled in L_1.

FIGURE 1. Physical dimensions - Continued.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-S-19500, and as specified herein.

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-S-19500.

4.3 Screening. Screening shall be in accordance with table II of MIL-S-19500, and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see table II of MIL-S-19500)	Measurements
	JANTX AND JANTXV levels
11	I_{CEX1} and h_{FE2}
12	See 4.3.1
13	Subgroup 2 of table I herein; Δh_{FE2} = ±40 percent I_{CEX1} = 100 percent of initial value or 100 nA dc whichever is greater.

4.3.1 Power burn-in conditions. Power burn-in conditions are as follows:

$V_{CE} \geq 10$ V dc; $T_J = 162.5°C \pm 12.5°C$

NOTE: No heat sink or forced air cooling on the devices shall be permitted.

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-S-19500.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-S-19500 and table I herein. Electrical measurements (end-points) and delta requirements shall be in accordance with the applicable steps of table II herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table IVb (JANTX and JANTXV) of MIL-S-19500. Electrical measurements (end-points) and delta requirements shall be in accordance with the applicable steps of table II herein.

4.4.2.1 Group B inspection, table IVb (JANTX and JANTXV) of MIL-S-19500.

Subgroup	Method	Conditions
B3	1037	$V_{CE} \geq 10$ V dc, 2,000 cycles.
B5	3151	$R_{\theta JC} = 2.33°C/W$.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table V of MIL-S-19500. Electrical measurements (end points) and delta requirements shall be in accordance with the applicable steps of table II herein.

4.4.3.1. Group C inspection, table V of MIL-S-19500.

Subgroup	Method	Conditions
C2	2036	Test condition A, weight = 10 lbs, t = 15 s.
C6	1037	$V_{CE} \geq 10$ V dc, 6,000 cycles.

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TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical examination	2071					
Subgroup 2						
Breakdown voltage, collector - emitter	3011	Bias condition. D; I_C = 100 mA dc; pulsed (see 4.5.1)	$V_{(BR)CEO}$			
2N6300				60		V dc
2N6301				80		V dc
Collector - emitter cutoff current	3041	Bias condition A; V_{BE} = -1.5 V dc;	I_{CEX1}			
2N6300		V_{CE} = 60 V dc			0.5	mA dc
2N6301		V_{CE} = 80 V dc			0.5	mA dc
Collector - emitter cutoff current	3041	Bias condition D;	I_{CEO}			
2N6300		V_{CE} = 30 V dc			0.5	mA dc
2N6301		V_{CE} = 40 V dc			0.5	mA dc
Emitter - base cutoff current	3061	Bias condition D; V_{EB} = 5 V dc	I_{EBO}		2.0	mA dc
Forward - current transfer ratio	3076	V_{CE} = 3 V dc; I_C = 1 A dc; pulsed (see 4.5.1)	h_{FE1}	500		
Forward - current transfer ratio	3076	V_{CE} = 3 V dc; I_C = 4 A dc; pulsed (see 4.5.1)	h_{FE2}	750	18000	
Forward - current transfer ratio	3076	V_{CE} = 3 V dc; I_C = 8 A dc; pulsed (see 4.5.1)	h_{FE3}	100		
Base - emitter voltage (nonsaturated)	3066	Test condition B; V_{CE} = 3 V dc; I_C = 4 A dc; pulsed (see 4.5.1)	$V_{BE(on)}$		2.8	V dc
Base - emitter voltage (saturated)	3066	Test condition A; I_C = 8 A dc; I_B = 80 mA dc; pulsed (see 4.5.1)	$V_{BE(sat)}$		4.0	V dc
Collector - emitter voltage (saturated)	3071	I_C = 4 A dc; I_B = 16 mA dc; pulsed (see 4.5.1)	$V_{CE(sat)1}$		2.0	V dc

See footnote at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 2 - cont.						
Collector - emitter voltage (saturated)	3071	I_C = 8 A dc; I_B = 80 mA dc; pulsed (see 4.5.1)	$V_{CE(sat)2}$		3.0	V dc
Subgroup 3						
High temperature operation		T_A = +150°C				
Collector - emitter cutoff current 2N6300 2N6301	3041	Bias condition A; V_{BE} = -1.5 V dc; V_{CE} = 60 V dc V_{CE} = 80 V dc	I_{CEX2}		 5.0 5.0	 mA dc mA dc
Low temperature operation		T_A = -55°C				
Forward - current transfer ratio	3076	V_{CE} = 3 V dc; I_C = 4 A dc; pulsed (see 4.5.1)	h_{FE4}	200		
Subgroup 4						
Small signal short circuit forward current transfer ratio	3206	V_{CE} = 3 V dc; I_C = 3 A dc; f = 1 kHz	h_{fe}	300		
Magnitude of small signal short - circuit forward - current transfer ratio	3306	V_{CE} = 3 V dc; I_C = 3 A dc; f = 1.0 MHz	$\lvert h_{fe} \rvert$	25	350	
Pulse response						
Turn-on time		(See figure 2); V_{CC} = 30 V dc; I_C = 4 A dc; I_{B1} = 16 mA dc	t_{on}		2.0	µs
Turn-off time		(See figure 3) V_{CC} = 30 V dc; I_C = 4 A dc; I_{B1} = I_{B2} = 16 mA dc	t_{off}		8.0	µs
Open circuit output capacitance	3236	V_{CB} = 10 V dc; I_E = 0; 100 kHz ≤ f ≤ 1 MHz	C_{obo}		200	pF

See footnote at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 5						
Safe operating area (DC)	3051	T_C = 25°C +10°C; t = 1s; 1 cycle; (see figure 4)				
Test 1		V_{CE} = 8 V dc; I_C = 8 A dc				
Test 2		V_{CE} = 30 V dc; I_C = 2.13 A dc				
Test 3 2N6300 2N6301		I_C = 100 mA dc; V_{CE} = 60 V dc V_{CE} = 80 V dc				
Safe operating area (switching)	3053	Load condition B; (clamped inductive load) T_A = 25°C; $t_r + t_f \leq 1.0$ µs duty cycle ≤ 10% t_p = 1 ms; (vary to obtain I_C); R_s = 0.1 ohms; R_{BB1} = 80 ohms; V_{BB1} = 16 V dc; R_{BB2} = 100 ohms; V_{BB2} = 1.5 V dc; V_{CC} = 50 V dc; I_C = 8 A dc; $R_L \leq 2$ ohms; L = 1 mH; clamp voltage =				
2N6300		60 V dc				
2N6301		80 V dc				
Safe operating area (switching)	3053	Load condition C; (unclamped inductive load) (see figure 5) T_A = 25°C; duty cycle ≤ 10%; $R_S \leq 0.1$ ohm				
Test 1		t_p = 1 ms (vary to obtain IC); R_{BB1} = 80 ohms; $V_{BB1} \geq 12$ V dc; $R_{BB2} = \infty$; $V_{CC} \geq 30$ V dc; I_C = 8 A dc; $R_L \leq 0.5$ ohms; L = 1 mH at 8 A dc				

See footnote at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 5 - cont.						
Test 2		t_p = 1 ms (vary to obtain IC); R_{BB1} = 80 ohms; V_{BB1} ≥ 12 V dc; R_{BB2} = ∞; V_{BB2} = 0 V dc; V_{CC} ≥ 30 V dc; I_C = 0.2 A dc; R_L ≤ 0.5 ohms; L = 100 mH at 0.2 A dc				
End point electrical measurements		See table II steps 1 and 3				
Subgroups 6 & 7						
Not applicable						

1/ For sampling plan see MIL-S-19500

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TABLE II. Groups B and C electrical measurements. 1/ 2/

Step	Inspection	MIL-STD-750		Symbol	Limit		Unit
		Method	Conditions		Min	Max	
1.	Collector - emitter cutoff current 2N6300 2N6301	3041	Bias condition A; V_{BE} = 1.5 V dc; V_{CE} = 60 V dc V_{CE} = 80 V dc	I_{CEX1}		0.5	mA dc
2.	Collector - emitter cutoff current 2N6300 2N6301	3041	Bias condition A; V_{BE} = 1.5 V dc; V_{CE} = 60 V dc V_{CE} = 80 V dc	I_{CEX2}		1.0	mA dc
3.	Forward - current transfer ratio	3076	V_{CE} = 3 V dc; I_C = 4 A dc pulsed (see 4.5.1)	h_{FE2}	750	18000	
4.	Forward - current transfer ratio	3076	V_{CE} = 3 V dc; I_C = 4 A dc pulsed (see 4.5.1)	Δh_{FE2}	±40%		

1/ The electrical measurements for table IVb (JANTX and JANTXV) of MIL-S-19500 are as follows:

a. Subgroup 2, see table II herein, steps 1 and 3.

b. Subgroup 3, see table II herein, steps 2 and 4.

c. Subgroup 6, see table II herein, steps 2 and 4.

2/ The electrical measurements for table V of MIL-S-19500 are as follows:

a. Subgroup 3, see table II herein, steps 1 and 3.

b. Subgroup 6, see table II herein, steps 2 and 4.

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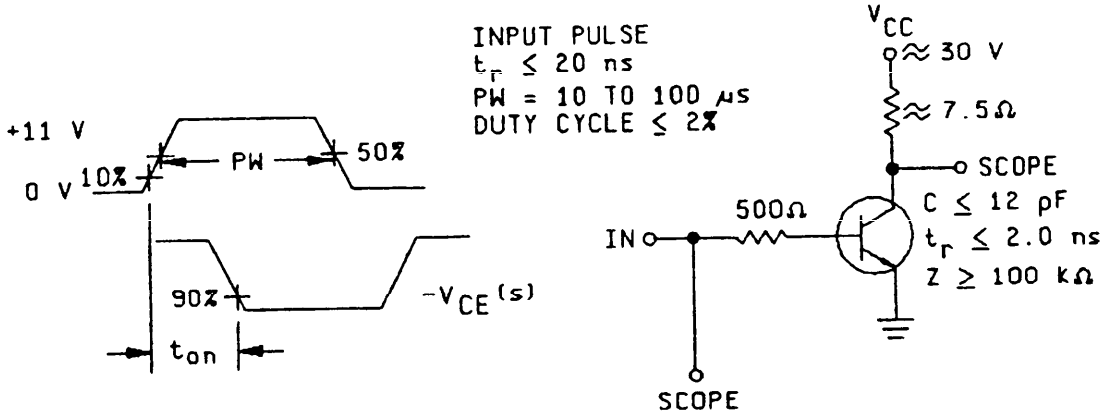


FIGURE 2. Turn on (t_{on}) time test circuit.

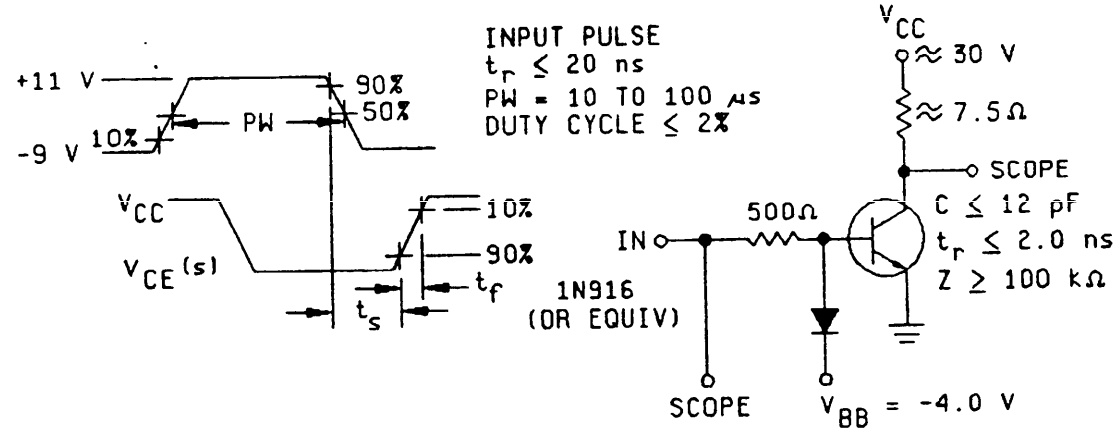


FIGURE 3. Switching time test circuits.

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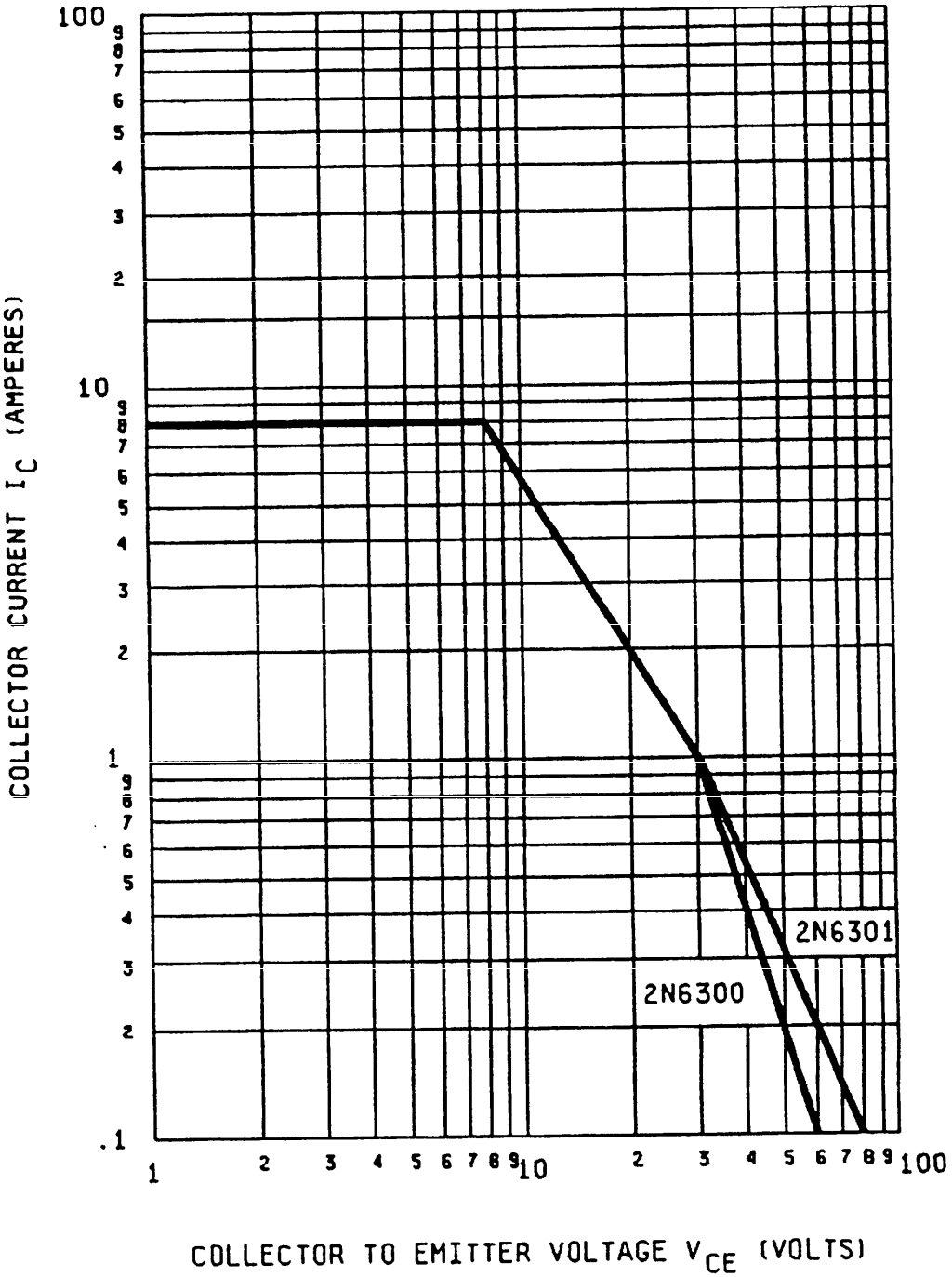


FIGURE 4. Maximum safe operating graph (dc).

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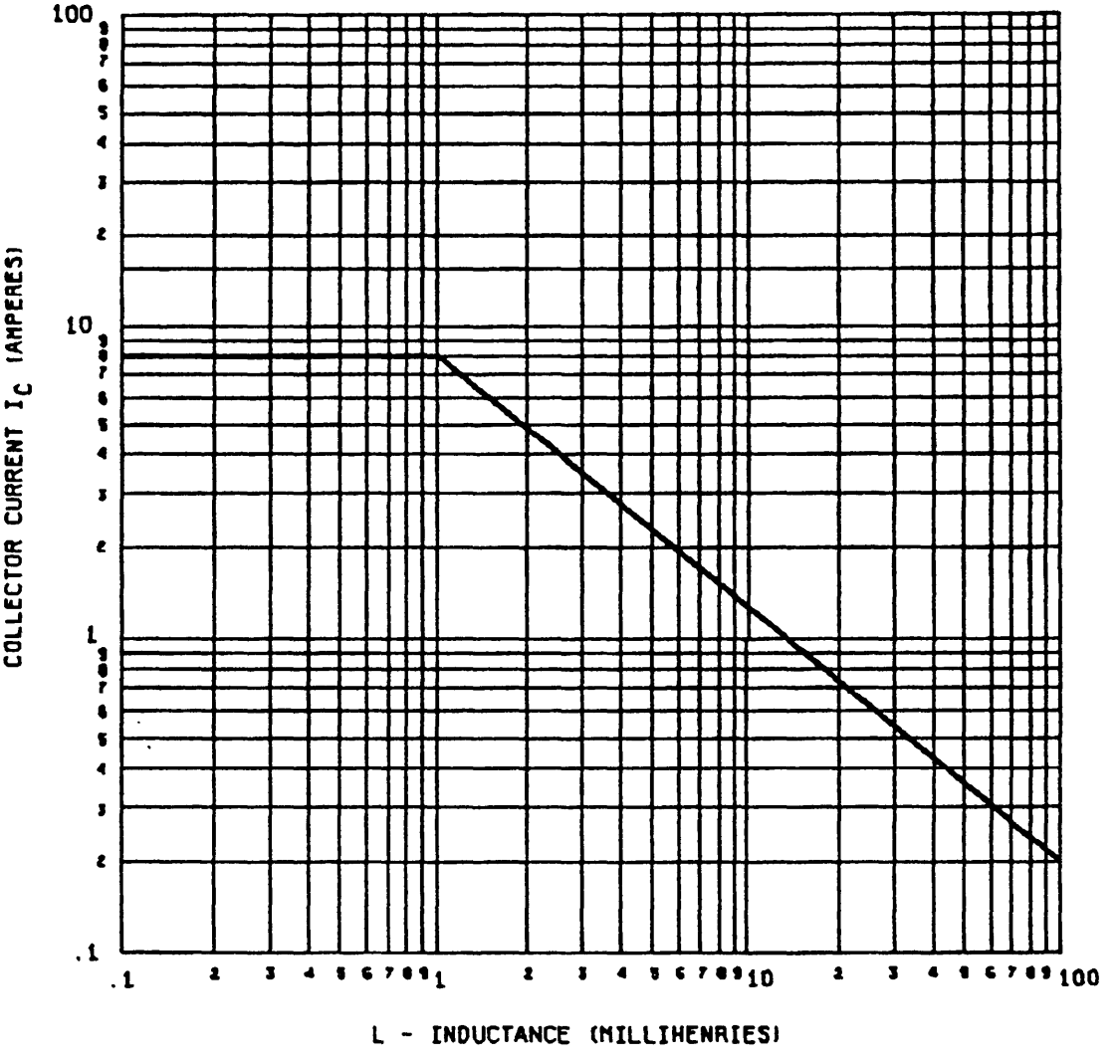


FIGURE 5. Safe operating area for switching between saturation and cutoff (unclamped inductive load).

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4.5 Method of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-S-19500.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Notes. The notes specified in MIL-S-19500 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Issue of DODISS to be cited in the solicition.

b. Lead finish as specified (see 3. 3.1).

c. Type designation and product assurance level.

6.3 Substitution of JAN devices. JANTX quality level devices are a oneway direct substitute for JAN quality level devices (example JANTX2N6300 for JAN2N6300). Manufacturers can no longer manufacture device types from this specification with a JAN assurance level.

6.4 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 17
NASA -NA

Review activities:
Air Force - 19, 80, 85,99
DLA - ES

Preparing activity:
NASA - NA

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-S-19500/539A	2. DOCUMENT DATE (YYMMDD)

3. DOCUMENT TITLE SEMICONDUCTOR DEVICE, DARLINGTON TRANSISTOR, NPN, SILICON, POWER TYPES 2N6300, 2N6301 JANTX AND JANTXV

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (if applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON

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